

From First Encounter to Chronic Care:

A Multi-Level Analysis of Patient Utilization, Access Gaps, and Social Determinants in a Regional Health System

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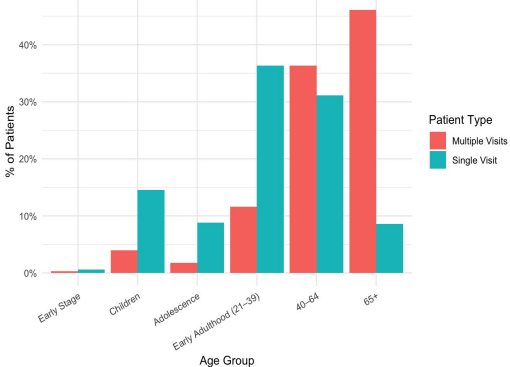
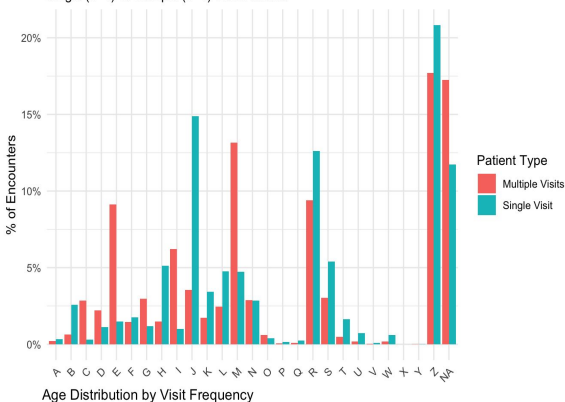
DataFest 2026



01 Patient Patterns: Who returns to Care?

Shawnee County (FIPS 201770) comprises ~48.9% of all multiple-visit encounters — the strongest proxy for longitudinal care behavior in the SVH system.

Distribution of Diagnosis Categories by Visit Frequency
Single (n=1) vs Multiple (n>1) Visit Patients

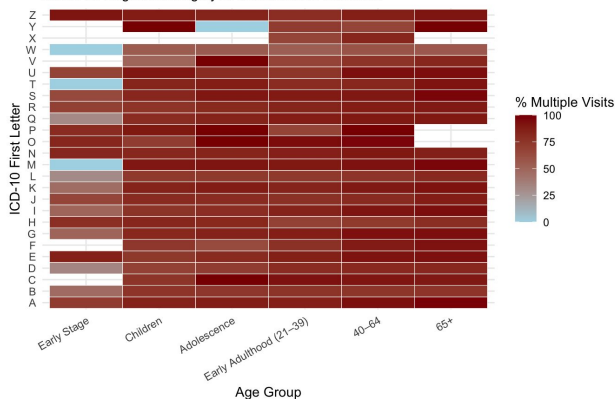


Likely-to-Return-to-Care Groups (≥10,000 Patients)

High-Volume Age × Diagnosis Combinations (n ≥ 10,000)			
Age Group	ICD-10 Chapter	% Multiple Visits	N Patients
65+	M	96.4	10532
65+	I	94.2	10729
40–64	M	92.8	13466
65+	Z	91.7	12248
65+	R	89.5	12328
Children	Z	88.5	13117
40–64	R	87.8	19543
40–64	Z	87.1	25582
Early Adulthood (21–39)	R	83.9	14086
Early Adulthood (21–39)	Z	79.3	18046

Risk of Becoming a Multiple-Visit Patient

Based on diagnosis category at first observed encounter



..... χ^2 confirmed: age group, diagnosis type, and admission type all significantly predict return ($p < 0.001$)

- Diagnosis differences:** Single-visit patients are more concentrated in routine (Z) and certain symptom categories, while multi-visit patients show stronger representation in chronic and follow-up-related diagnoses.
- Age differences:** Older patients (40–64, 65+) are much more likely to have multiple visits, while younger patients are more likely to have single encounters.
- Notable Patterns:**
 - 65+ patients with M or I codes return at 94–96%.
 - Children show 88.5% return on Z codes — routine pediatric care is already longitudinal by nature.
 - Young adults (21–39) with R codes return at 83.9% but represent only 12% of multi-visit share — the highest-upside conversion target.

02 Geospatial Mapping + Provider Gaps

Methodology:

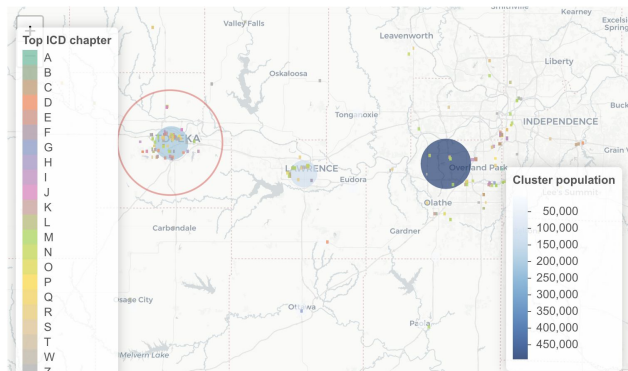
Patient Mapping
(FIPS codes + Census Centroids,
Bubble Size = Population)

Provider Geocoding
(Addresses + Census API Coordinates,
Mapped by Specialty Type)

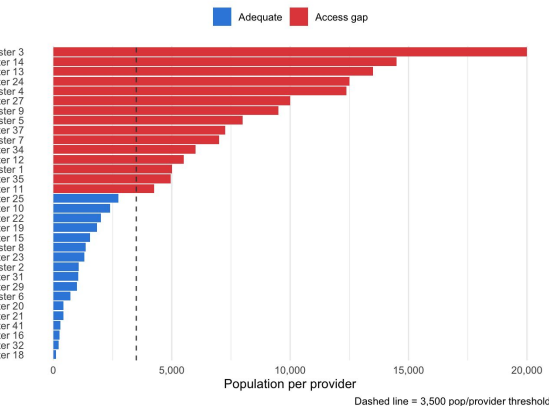
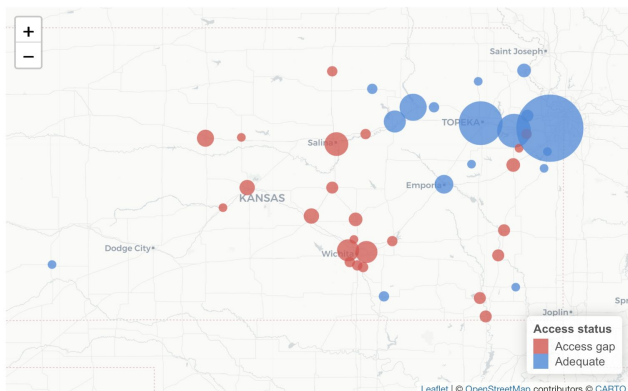
DBSCAN Clustering
(Population Weighted Clustering
Identifies Demand Zones)

Access Gap Flag
(Clusters Exceeding 3500 pop /
provider = gap by HPSA standard)

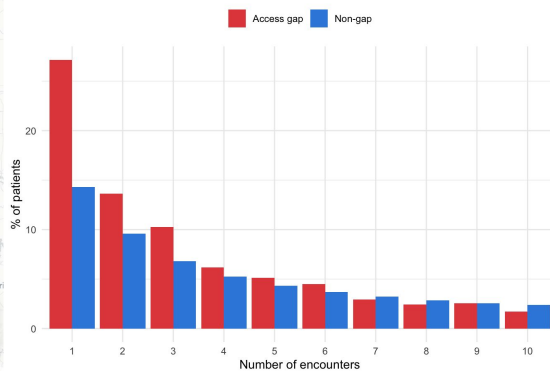
Cluster Summary Map



Access Gap Cluster Map



Encounter Frequency Distribution



% of Chronic Patients Who Are Single-Visit

Chronic Patients Who Appear Only Once

Group	Total Patients	Single-Visit	% Single-Visit
Access gap	1246	496	39.8
Non-gap	226750	49270	21.7

Access the Clustering and GeoSpatial Mapping [Here](#).

Findings:

- **24** clusters flagged as access gaps
- **6** clusters have zero providers
- **20,000:1** worst population-to-provider ratio
- **39.8%** of chronic gap-area patients appear only once vs. 21.7% non-gap
- **+9 days** longer average care journey gap vs. non-gap patients

Minimum Recommended Pathway:

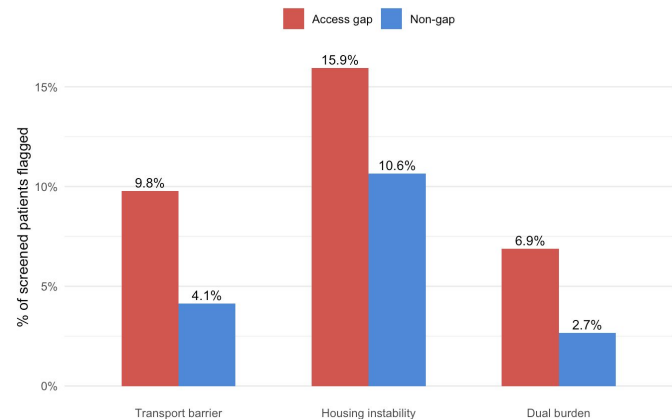
1. Deploy mobile or satellite clinic units to top 5 gap clusters (3, 14, 13, 24, 4)
2. Establish telehealth referral pathways for E and M chapter patients in zero-provider clusters
3. Priority: the 6 clusters with no providers at all

03 SDOH as a Fragility Signal + Recommendations

SDOH Screener Coverage by Access Group

Group	Total Patients	Screened	Coverage (%)
Access gap	1,929	276	14.3%
Non-gap	357,831	71,611	20%

SDOH Flag Rates: Access Gap vs. Non-Gap (Screened Patients Only)
Restricted to patients with at least one SDOH screening record



**18.8% – Any SDOH Flag
(Gap Patients)**

**28.6% – Gap + SDOH with
> 180d Journey Gap**

**6.9% – Dual Burden
(Gap Patients)**

**28.8% – Gap + SDOH who
are single-visit**

Screener data concentrated in Shawnee Co. — SDOH flags used as patient-level fragility indicators, not geographic prevalence estimates.

Sustaining & Growing Longitudinal Care:

1. **Convert high-return demographics:** Deepen continuity with 40–64 and 65+ cohorts presenting with M, I, E diagnoses. Already at 92–96% return rates.
2. **Engage young adult single-visit patients:** 83.9% return when presenting with symptoms. Post-visit follow-up protocols can convert acute visitors to longitudinal attenders.
3. **Routine care as a retention anchor:** Z-code visits carry high return rates across all age groups. Position annual wellness visits as longitudinal touchpoints.
4. **SDOH-informed care planning:** For transport/housing-flagged patients, offer telehealth follow-up and schedule the next appointment before discharge.

Addressing access gaps:

- **Minimum** — **Mobile & telehealth** — Deploy to Clusters 3, 14, 13. Telehealth referral network for 6 zero-provider clusters. Target E and M chapter chronic management.
- **Optimal** — **Provider recruitment** — Rural incentive programs for 1–2 PCPs per critical gap cluster. Co-locate chronic care and pharmacy given high E/M burden.
- **Care navigator role** — Gap patients average 172 days between visits. A dedicated navigator could close this and reduce the 39.8% single-visit chronic rate.
- **Expand SDOH screening** — Only 14.3% of gap-area patients are currently screened. Expansion would surface the ~18.8% carrying transport or housing barriers who need targeted support.



Thank you!

- Jellyfish -